

[illegible]
$$\begin{aligned}
Y_1 &= 3 \binom{(4)}{\kappa_1} - 4 \binom{(4)}{\kappa_{15}} + 3 \binom{(4)}{\kappa_2} + 2 \binom{(4)}{\kappa_3} \&\& Y_2 = 4 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_4} \&\& \\
Y_3 &= 2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4} \&\& Y_4 = \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} \&\& \text{Det}(1^+) = \\
&-\frac{1}{8} k^2 (k^2 (12 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_2} + 6 \binom{(4)}{\kappa_2} \binom{(4)}{\kappa_3} + 4 \binom{(4)}{\kappa_3}^2 + 6 \binom{(4)}{\kappa_2} \binom{(4)}{\kappa_4} + 4 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4} + \binom{(4)}{\kappa_4}^2 + 6 \binom{(4)}{\kappa_1} (2 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})) + \\
&12 (\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2}) \binom{(2)}{\kappa_1}) \&\& \text{Det}(2^+) = \frac{3}{2} k^2 (\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2})
\end{aligned}$$
$$\begin{aligned} & \overset{(2)}{\kappa}_1 \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(2)}{\kappa}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \overset{(4)}{\kappa}_2 \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta + \\ & \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - 3 \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - 3 \overset{(4)}{\kappa}_2 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta - \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta + \\ & \frac{1}{2} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \frac{1}{2} \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$